

**Directions:** Read all the questions carefully and answer them completely. No credit will be given for undocumented responses. **Show all your work to get full credit.** Give **exact answers** to questions unless explicitly requested otherwise.

1. Find the equilibrium temperature distribution, in the case where there is uniform heating along the length of the rod,

$$u_t = u_{xx} + 1, \quad u(x, 0) = f(x), \quad u_x(0, t) = 4, \quad u(2, t) = 5.$$

2. Show that is  $L(u) = k \frac{\partial^2 u}{\partial x^2} + \alpha(x, t)u$  is a linear operator.

3. Use the trigonometric identity

$$\cos a \cos b = \frac{1}{2} [\cos(a - b) + \cos(a + b)]$$

to evaluate  $\int_0^L \cos(n\pi x/L) \cos(m\pi x/L), n \geq 0, m \geq 0$ . (Be careful if  $n = m$ )

4. Consider the heat equation

$$\begin{aligned} \frac{\partial u}{\partial t} &= k \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < 3, \quad t > 0, \\ \frac{\partial u}{\partial x}(0, t) &= 0, \quad u(3, t) = 0, \quad t > 0, \\ u(x, 0) &= 5 \cos(\pi x/6) - 7 \cos(5\pi x/6), \quad 0 < x < 3. \end{aligned}$$

**Derive** the associated eigenvalues and eigenfunctions.

5. Solve the eigenvalue problem

$$\begin{aligned} \frac{d^2 \phi}{dx^2} &= -\lambda \phi, \quad 0 < x < 2\pi, \\ \phi(0) &= \phi(2\pi), \quad \frac{d\phi}{dx}(0) = \frac{d\phi}{dx}(2\pi). \end{aligned}$$